Sean Grate

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RESEARCH INTERESTS

Commutative algebra and its interactions with algebraic geometry and enumerative combinatorics: free resolutions and Betti numbers, Lefschetz properties, Castelnuovo–Mumford regularity, toric varieties, and matroids.

EDUCATION

2020–2025 **Ph.D. in Mathematics**
Auburn University · Advisor: Hal Schenck
Dissertation: *Problems in Computational Algebraic Geometry: Lefschetz Properties and Toric Varieties*

2016–2020 **B.S. in Mathematics, *cum laude***
University of Kentucky · Minors in History and Classics

APPOINTMENTS

2025–present **Postdoctoral Research Associate**
Iowa State University, Department of Mathematics
Mentor: Jason McCullough

2020–2025 **Graduate Teaching Assistant**
Auburn University, Department of Mathematics and Statistics

2019–2020 **Undergraduate Research Assistant**
University of Kentucky, Department of Computer Science
Machine learning for airborne LiDAR · Advisor: Nathan Jacobs

PUBLICATIONS

5. Haile Gilroy, Melinda Lanius, and **Sean Grate**. *Graph theoretic reflection to foster alignment in coordinated courses*. International Journal of Mathematical Education in Science and Technology (2025). Published online., 1–18.
4. Ayah Almousa, **Sean Grate**, Daoji Huang, Patricia Klein, Adam LaClair, Yuyuan Luo, and Joseph McDonough. *The MatrixSchubert package for Macaulay2*. J. Softw. Algebra Geom. 15.1 (2025), 41–55.
3. Nasrin Altafi, Roberta Di Gennaro, Federico Galetto, **Sean Grate**, Rosa M. Miró-Roig, Uwe Nagel, Alexandra Seceleanu, and Junzo Watanabe. *Betti numbers for connected sums of graded Gorenstein Artinian algebras*. Trans. Amer. Math. Soc. 378.2 (2025), 1055–1080.
2. **Sean Grate** and Hal Schenck. *Betti tables forcing failure of the weak Lefschetz property*. Lefschetz properties—current and new directions. **59**. Springer INdAM Ser. Springer, Singapore, 2024, 155–164.
1. Hunter Blanton, **Sean Grate**, and Nathan Jacobs. *Surface Modeling for Airborne Lidar*. IGARSS 2020 – IEEE International Geoscience and Remote Sensing Symposium. 2020, 1110–1113.

PREPRINTS

3. Herman Chau, Spencer Daugherty, **Sean Grate**, Mark D. Haiman, Juan Carlos Martínez Mori, and Anna Pun. *The Stable Tamari Lattice*. 2026. arXiv:2610.39257.
2. Louiza Fouli, **Sean Grate**, Selvi Kara, Adam LaClair, Jason McCullough, Vinh Nguyen, Aleksandra Sobieska, and Prajwal Udanshive. *Graded Betti numbers of graded Möbius algebras of uniform matroids*. Submitted. 2026. arXiv:2609.03827.
1. Barbara Betti, **Sean Grate**, Thiago Holleben, and Flavio Salizzoni. *Incidence toric ideals and three-point functions*. Submitted. 2026. arXiv:2605.23849.

IN PREPARATION

1. **Sean Grate** and Jason McCullough. *Gorenstein graded Möbius algebras are Koszul.* 2026.

GRANTS

2026 **NSF conference grant DMS-2625606** · $35,000
John Cobb (PI), Michael Brown (Co-PI), **Sean Grate** (Co-PI)
Supports a workshop at Auburn University for development of software for Macaulay2.

SOFTWARE

2025 **EquivariantEhrhart** · Macaulay2 package for equivariant Ehrhart theory.
In development.

2023 **Permutations** · Macaulay2 package providing a permutation type.
Distributed with Macaulay2.

2023 **MatrixSchubert** · Macaulay2 package for matrix Schubert and ASM varieties.
Distributed with Macaulay2; documented in *J. Softw. Algebra Geom.* **15** (2025).

Languages and tools · Macaulay2, Python, SageMath, PyTorch, Git, LaTeX

AWARDS AND HONORS

2024 Baskervill Endowed Mathematics Fellowship, Auburn University DMS
2023 Ralph B. Bennett Memorial Endowed Mathematics Fellowship, Auburn University DMS
2022, 2024 Teaching Citation Award, Auburn University DMS
2021, 2022 Research Citation Award, Auburn University DMS
2022 Outstanding Graduate Teaching Assistant Award, Auburn University COSAM
2016–2020 Provost Scholarship, University of Kentucky · $1,500 per year
2016–2020 Kentucky Educational Excellence Scholarship · $2,225 per year
2016–2020 Dean's List, University of Kentucky · multiple semesters

INVITED CONFERENCE TALKS

15. *TBD*
April 2027 · AMS Spring Central Sectional Meeting (upcoming)
14. *TBD*
April 2027 · CA+ (upcoming)

13. *Incidence toric ideals and three-point functions*
January 2027 · Joint Mathematics Meetings (upcoming)
12. *Gorenstein graded Möbius algebras are Koszul*
January 2027 · Joint Mathematics Meetings (upcoming)
11. *3-point functions and incidence toric ideals*
October 2026 · SIAM Texas–Louisiana Sectional Meeting
10. *The graded Möbius algebra of a matroid*
August 2026 · Upcoming Researchers in Commutative Algebra (URiCA)
9. *Gorenstein graded Möbius algebras are Koszul*
April 2026 · AMS Spring Central Sectional Meeting
8. *Combinatorial bounds on the Castelnuovo–Mumford regularity of toric surfaces*
October 2025 · AMS Fall Central Sectional Meeting
7. *Combinatorial bounds on the Castelnuovo–Mumford regularity of toric surfaces*
July 2025 · Mathematical Congress of the Americas
6. *Castelnuovo–Mumford regularity of toric surfaces*
July 2025 · SIAM Conference on Applied Algebraic Geometry
5. *Combinatorial bounds on the Castelnuovo–Mumford regularity of toric surfaces*
May 2025 · AWM Research Symposium
4. *Castelnuovo–Mumford regularity of toric surfaces*
January 2025 · Joint Mathematics Meetings
3. *Castelnuovo–Mumford regularity of toric surfaces*
September 2024 · AMS Fall Central Sectional Meeting
2. *Betti tables forcing failure of the weak Lefschetz property*
January 2024 · Combinatorial Algebra meets Algebraic Combinatorics (CAAC)
1. *Betti tables forcing failure of the weak Lefschetz property*
May 2023 · Workshop on Lefschetz Properties in Algebra, Geometry, Topology and Combinatorics

INVITED SEMINAR TALKS

13. *TBD*
· Algebra Seminar, University of Iowa (upcoming)
12. *Gorenstein graded Möbius algebras are Koszul*
November 2026 · Commutative Algebra Seminar, University of Nebraska–Lincoln (upcoming)
11. *The stable Tamari lattice*
November 2026 · Discrete Mathematics Seminar, University of Nebraska–Lincoln (upcoming)
10. *The graded Möbius algebra of a matroid*
October 2026 · Algebra Seminar, New Mexico State University
9. *SVD is all you need*
May 2026 · Undergraduate Seminar, Grinnell College
8. *Betti tables forcing failure of the weak Lefschetz property*
November 2025 · Stockholm Commutative Algebra Seminar (virtual)
7. *Combinatorial bounds on the Castelnuovo–Mumford regularity of toric surfaces*
February 2025 · Syzygies and Mirror Symmetry Seminar (virtual)
6. *Betti tables forcing failure of the weak Lefschetz property*
October 2024 · Commutative Algebra Seminar, University of Nebraska–Lincoln

5. *Betti tables and Lefschetz properties*
   October 2024 · Commutative Algebra Reading Seminar, University of Nebraska–Lincoln
4. *Betti numbers of connected sums of graded Artinian Gorenstein algebras*
   October 2024 · Commutative Algebra Seminar, Purdue University
3. *Betti tables forcing failure of the weak Lefschetz property*
   October 2024 · Student Commutative Algebra Seminar, Purdue University
2. *Betti tables and Lefschetz properties*
   November 2023 · Algebra Seminar, University of Kentucky
1. *Lefschetz properties and Artinian rings*
   November 2023 · Mathematics Seminar, McNeese State University

POSTER PRESENTATIONS

5. *Castelnuovo–Mumford regularity of toric surfaces*
   April 2025 · Commutative Algebra in the South (CATS), Georgia State University
4. *Castelnuovo–Mumford regularity of toric surfaces*
   March 2025 · Mid-Atlantic Algebra, Geometry, and Combinatorics (MAAGC), CUNY Graduate Center
3. *Castelnuovo–Mumford regularity of toric surfaces*
   January 2025 · Combinatorial Algebra meets Algebraic Combinatorics (CAAC), Fields Institute
2. *Betti numbers of connected sums of graded Artinian Gorenstein algebras*
   August 2024 · UweFest, University of Notre Dame
1. *Betti tables forcing failure of the weak Lefschetz property*
   August 2023 · BrianFest, University of Nebraska–Lincoln

CONFERENCE AND WORKSHOP ORGANIZATION

Apr 2027 **AMS Special Session on Advances in Commutative Algebra and Combinatorics** · University of Iowa, IA (upcoming)
Co-organizer with Adam LaClair and Jason McCullough.

Feb 2027 **Macaulay2 Workshop** · Auburn University, AL (upcoming)
Co-organizer with Michael Brown and John Cobb. Supported by NSF DMS-2625606.

Oct 2025 **AMS Special Session on Algebraic Combinatorics and Combinatorial Commutative Algebra** · Tulane University, LA
Co-organizer with Thiago Holleben and Anton Dochtermann.

MENTORING

Spring 2026 **Simplicial complexes forcing failure of the weak Lefschetz property**
Nick Smith and Deepak Woli · ISMaRT program, Iowa State University
Determining when an Artinian algebra associated to a simplicial complex has the weak Lefschetz property.

TEACHING

IOWA STATE UNIVERSITY

Spring 2027 **Recitation instructor** · MATH 1660, Calculus II (upcoming)

Fall 2026 **Instructor of record** · MATH 1500, Discrete Mathematics for Business and Social Sciences

Spring 2026 **Instructor of record** · MATH 2070, Matrices and Linear Algebra

Fall 2025 **Recitation instructor** · MATH 1660, Calculus II

AUBURN UNIVERSITY

2024 **Recitation leader** · MATH 1620, Calculus II · Spring and Fall

2022–2023 **Recitation leader** · MATH 1610, Calculus I · Fall 2022, Spring 2023, Fall 2023

Spring 2022 **Instructor of record** · MATH 1610, Calculus I

Fall 2021 **Instructor of record** · MATH 1120, Precalculus Algebra

SERVICE

2023–present **Referee** · *Journal of Algebra*; *Combinatorial Theory*; *Journal of Software for Algebra and Geometry* · technical reviewer, $\times 2$

2021–2024 **President** · Mathematics Club, Auburn University

2022–2023 **Secretary** · Graduate Student Council, Auburn University DMS

2021–2023 **Graduate student representative** · Graduate Student Council, Auburn University DMS

OUTREACH

2022, 2023 **Destination STEM**, Auburn University · interactive exhibits for Alabama schools

2022 **Auburn Mathematical Puzzle Challenge (AMP'd)** · team problem-solving competition

2019 **Expanding Your Horizons**, workshop mentor · hands-on mathematics demonstrations

2019 **Julia Robinson Mathematics Festival**, volunteer · open-ended mathematical puzzles